Lab Programs

SET 13- Geographic Data

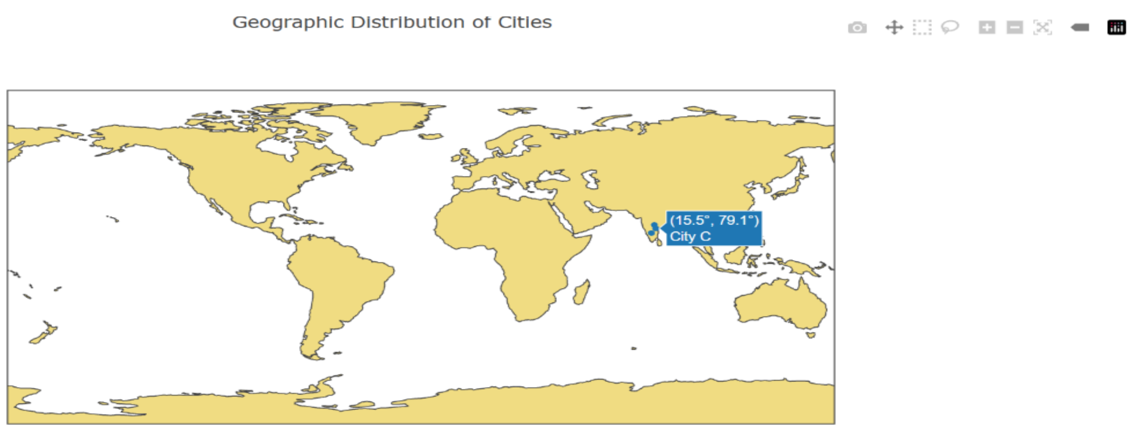
13A. Map Chart

Program:

```
install.packages("ggplot2")
install.packages("plotly")
install.packages("knitr")
library(ggplot2)
library(plotly)
library(knitr)
geo_data <- data.frame(
City=c("City A","City B","City C"),
Population=c(500000,700000,600000),
AvgTemperature=c(75,68,80),
Elevation=c(1000,800,1200),
Longitude=c(78.5,77.2,79.1),
Latitude=c(17.4,13.0,15.5))
plot_ly(
geo_data,
type="scattergeo",
lon=~Longitude,
lat=~Latitude,
text=~City,
mode="markers"
) %>%
layout(
title="Geographic Distribution of Cities",
```

geo=list(showland=TRUE))

Output:

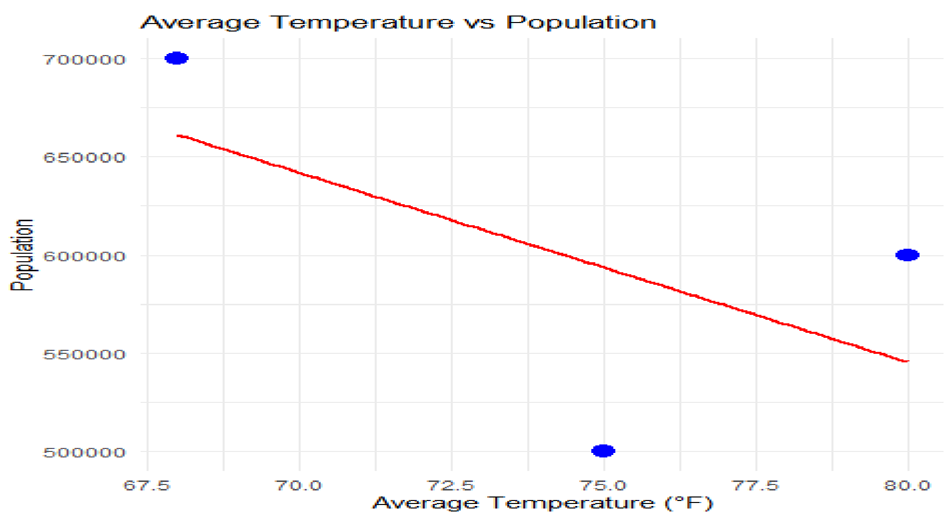


13B. Scatter Plot

Program:

```
ggplot(geo_data,
aes(x=AvgTemperature,
y=Population))+
geom_point(size=4,color="blue")+
geom_smooth(method="lm",se=FALSE,color="red")+
labs(title="Average Temperature vs Population",
x="Average Temperature (°F)",
y="Population")+
theme_minimal()
```

Output:



13C. Table

Program:

kable(

geo_data,

caption="Geographic Data of Cities"

)

Output:

```
Table: Geographic Data of Cities

|City    | Population| AvgTemperature| Elevation| Longitude| Latitude|
|:-------|----------:|--------------:|---------:|---------:|--------:|
|City A  |      5e+05|             75|      1000|      78.5|     17.4|
|City B  |      7e+05|             68|       800|      77.2|     13.0|
|City C  |      6e+05|             80|      1200|      79.1|     15.5|
>
```

13D. Interactive Dashboard

Program:

htmlwidgets::saveWidget(

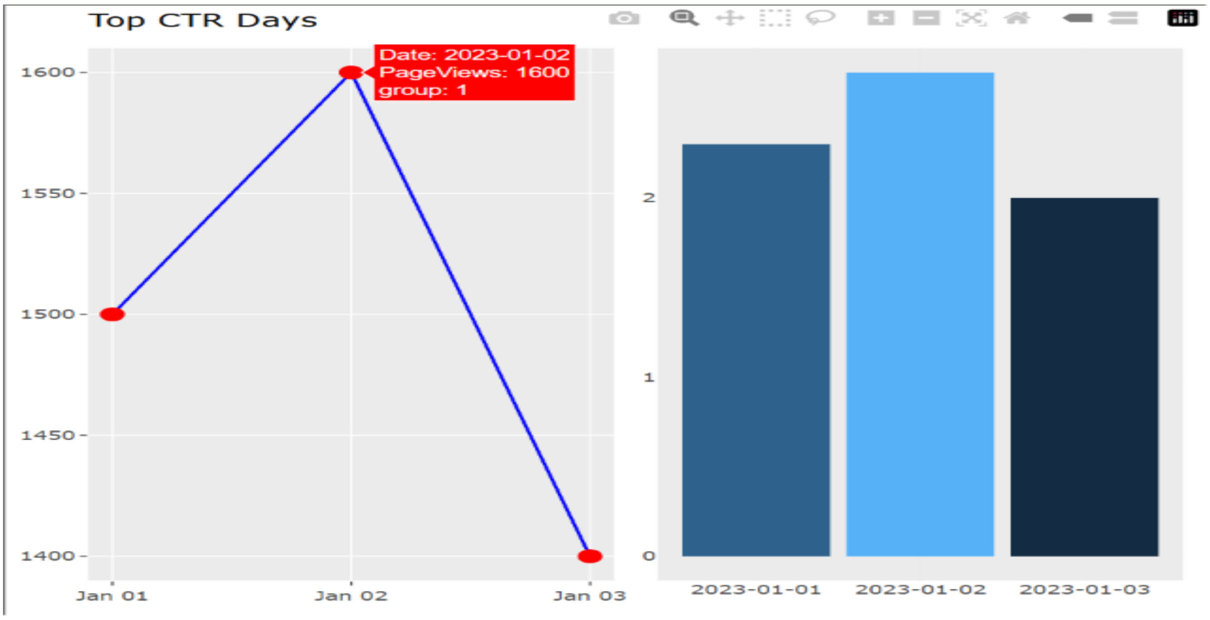
dashboard,

"GeographicDashboard.html",

selfcontained=TRUE)

browseURL("GeographicDashboard.html")

Output:



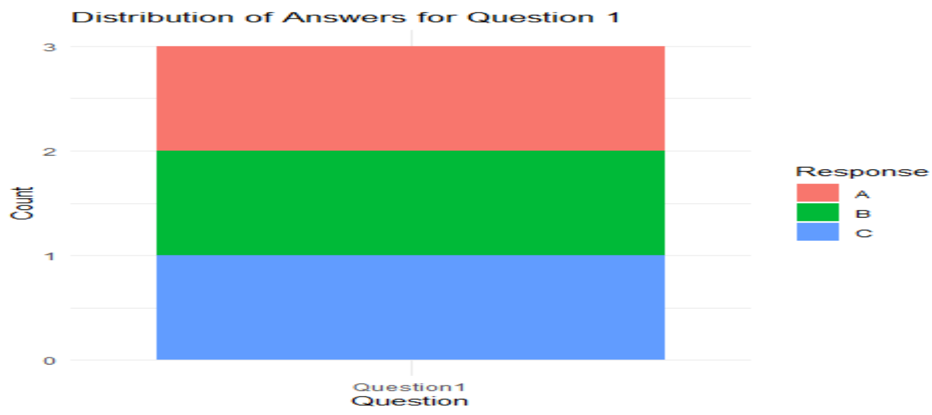
SET 14- Survey Results

14A. Stacked Bar Chart

Program:

```
install.packages("ggplot2")
install.packages("plotly")
install.packages("knitr")
install.packages("fmsb")
install.packages("reshape2")
survey_data <- data.frame(
Respondent=c(1,2,3),
Question1=c("A","B","C"),
Question2=c("B","A","A"),
Question3=c("C","D","B"))
survey_long <- melt(
survey_data,
id.vars="Respondent",
variable.name="Question",
value.name="Response")
ggplot(
survey_long[survey_long$Question=="Question1",],
aes(x=Question,fill=Response)
)+
geom_bar()+
labs(
title="Distribution of Answers for Question 1",
x="Question",
y="Count",
fill="Response"
)+
theme_minimal()
```

Output:

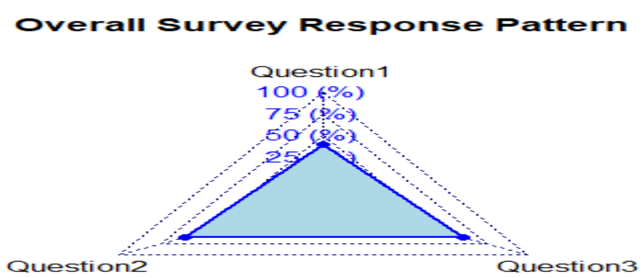


14B. Radar Chart

Program:

```
response_data <- data.frame(
Question=c("Question1","Question2","Question3"),
Score=c(2,3,3))
radar <- data.frame(
Max=c(5,5,5),
Min=c(0,0,0),
Average=response_data$Score)
colnames(radar)=c("Question1","Question2","Question3")
radarchart(radar,axistype=1,
pcol="blue",
pfcol="lightblue",plwd=2,
title="Overall Survey Response Pattern")
```

Output:



14C. Table

Program:

kable(

survey_data,

caption="Survey Response Data"

)

Output:

```
Table: Survey Response Data

| Respondent|Question1 |Question2 |Question3 |
|----------:|:---------|:---------|:---------|
|          1|A         |B         |C         |
|          2|B         |A         |D         |
|          3|C         |A         |B         |
> |
```

14D. Interactive Dashboard

Program:

htmlwidgets::saveWidget(

dashboard,

"SurveyDashboard.html",

selfcontained=TRUE)

browseURL("SurveyDashboard.html")

Output:

Show 10 entries

Search:

Survey Responses

	Respondent	Question1	Question2	Question3
1	1	A	B	C
2	2	B	A	D
3	3	C	A	B

Showing 1 to 3 of 3 entries

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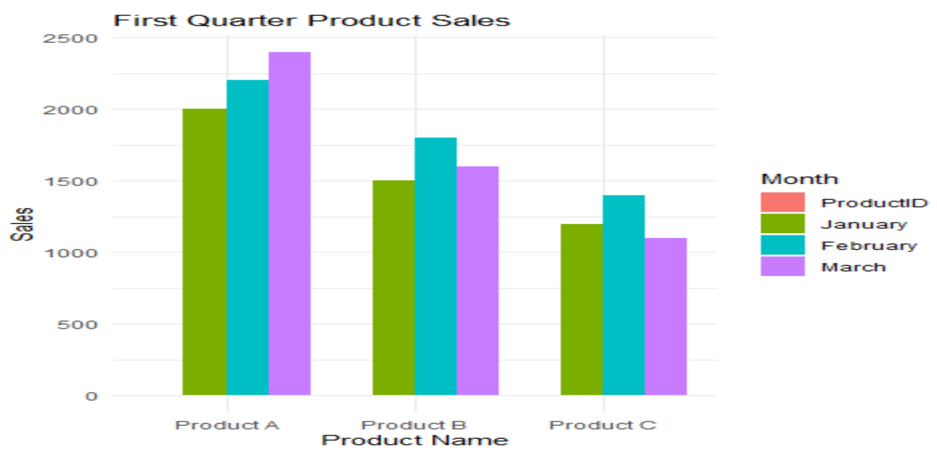
SET 15- Product Sales Analysis

15A. Grouped Bar Chart

Program:

```
sales_data <- data.frame(
ProductID=c(1,2,3),
ProductName=c("Product A","Product B","Product C"),
January=c(2000,1500,1200),
February=c(2200,1800,1400),
March=c(2400,1600,1100))
sales_long <- melt(sales_data,
id.vars="ProductName",
variable.name="Month",
value.name="Sales")
ggplot(sales_long,
aes(x=ProductName,
y=Sales,
fill=Month))+
geom_bar(stat="identity",position="dodge")+
labs(title="First Quarter Product Sales",
x="Product Name",
y="Sales")+
theme_minimal()
```

Output:

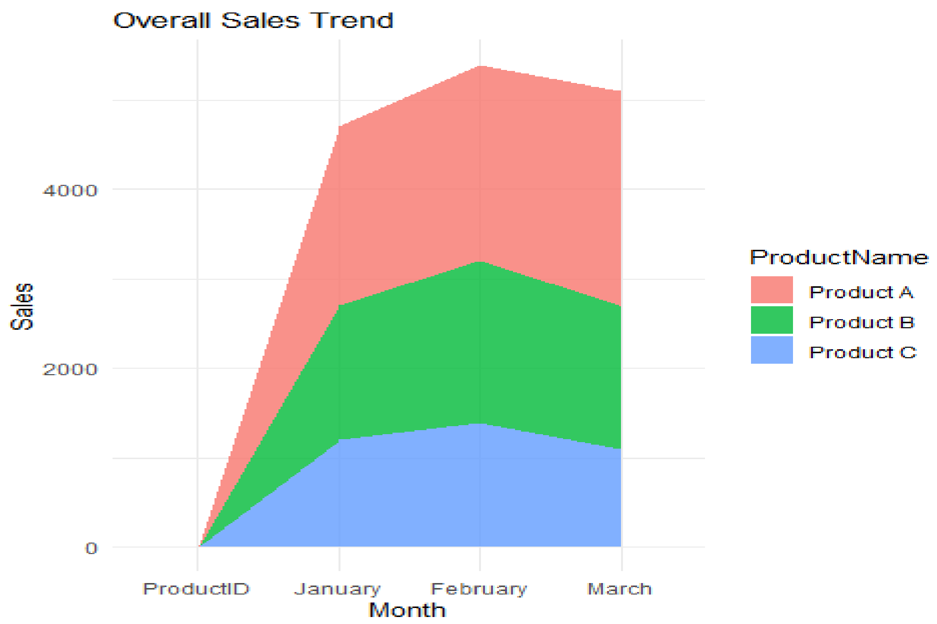


15B. Stacked Area Chart

Program:

```
ggplot(sales_long,
aes(x=Month,
y=Sales,
fill=ProductName,
group=ProductName))+
geom_area(alpha=0.8)+
labs(title="Overall Sales Trend",
x="Month",
y="Sales")+
theme_minimal()
```

Output:



15C. Interactive Table

Program:

```
datatable(
sales_data,
caption="Monthly Product Sales",
options=list(pageLength=10)
)
```

Output:

Show 10 entries

Search:

Monthly Product Sales

	ProductID	ProductName	January	February	March
1	1	Product A	2000	2200	2400
2	2	Product B	1500	1800	1600
3	3	Product C	1200	1400	1100

Showing 1 to 3 of 3 entries

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15D. 4: Interactive Dashboard

Program:

```
library(plotly)
bar_plot <- ggplotly(
ggplot(sales_long,
aes(x=ProductName,
y=Sales,
fill=Month))+
geom_bar(stat="identity",position="dodge")+
labs(title="Grouped Bar Chart")+
theme_minimal())
area_plot <- ggplotly(
ggplot(sales_long,
aes(x=Month,
y=Sales,
fill=ProductName,
group=ProductName))+
geom_area(alpha=0.8)+
labs(title="Stacked Area Chart")+
theme_minimal())
dashboard <- subplot(
bar_plot,
area_plot,
nrows=1,
```

```
shareX=FALSE,
shareY=FALSE)
dashboard
```

Output:

Monthly_Product_Sales.csv × Gemi

1 to 3 of 3 entries Filter

Product	January	February	March
Product A	2000	2200	2400
Product B	1500	1800	1600
Product C	1200	1400	1100

Show 10 per page

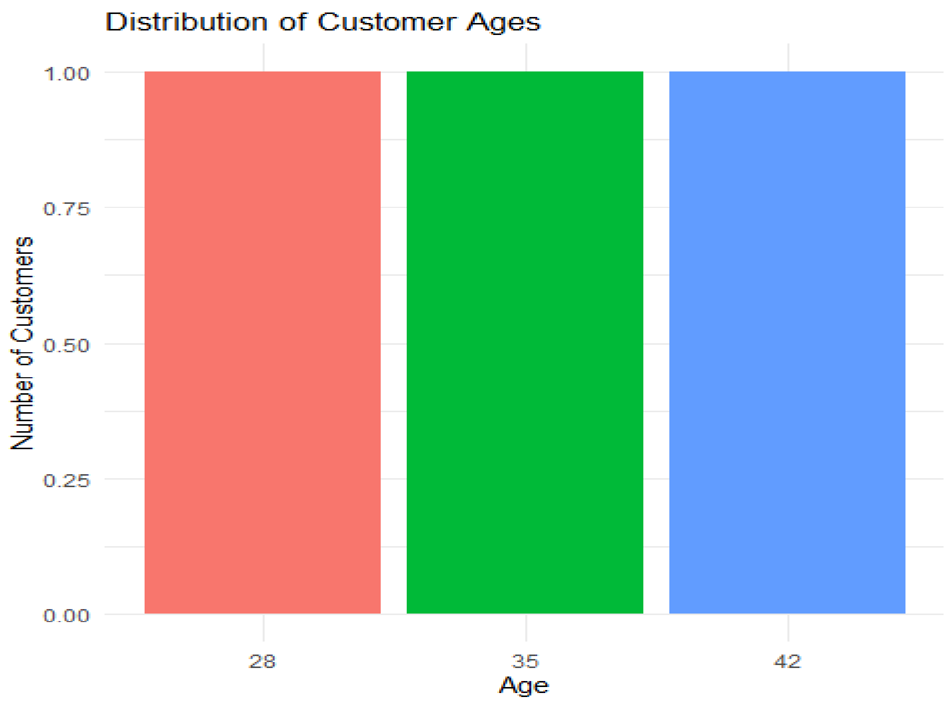
SET 16- Customer Demographics Analysis

16A. Line Chart

Program:

```
library(ggplot2)
customer_data <- data.frame(
  CustomerID = c(1,2,3),
  Age = c(28,35,42),
  Gender = c("Female","Male","Female"),
  Income = c(50000,60000,75000)
)
ggplot(customer_data, aes(x = factor(Age), fill = factor(Age))) +
  geom_bar() +
  labs(
    title = "Distribution of Customer Ages",
    x = "Age",
    y = "Number of Customers"
  ) +
  theme_minimal() +
  theme(legend.position = "none")
```

Output:

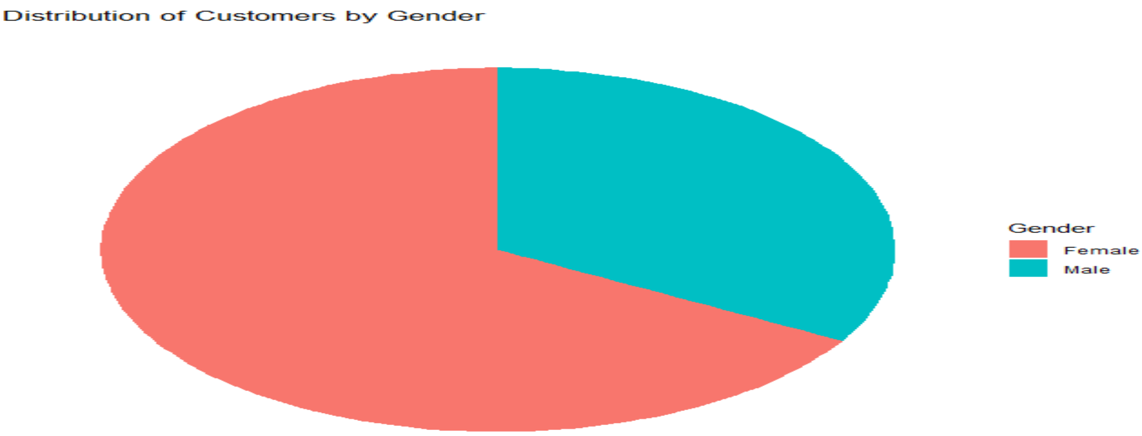


16B. Bar Chart

Program:

```
library(ggplot2)
customer_data <- data.frame(
  CustomerID = c(1,2,3),
  Age = c(28,35,42),
  Gender = c("Female","Male","Female"),
  Income = c(50000,60000,75000)
)
gender_data <- as.data.frame(table(customer_data$Gender))
colnames(gender_data) <- c("Gender","Count")
ggplot(gender_data, aes(x = "", y = Count, fill = Gender)) +
  geom_bar(stat = "identity", width = 1) +
  coord_polar(theta = "y") +
  labs(title = "Distribution of Customers by Gender") +
  theme_void()
```

Output:



16C. Table

Program:

View(customer_data)

Output:

Data: customer_data

	CustomerID	Age	Gender	Income
1	1	28	Female	50000
2	2	35	Male	60000
3	3	42	Female	75000

16D. Tableau Dashboard

Program:

CustomerID,Age,Gender,Income

1,28,Female,50000

2,35,Male,60000

3,42,Female,75000

Output:

CustomerI	Age	Gender	Income
1	28	Female	50000
2	35	Male	60000
3	42	Female	75000

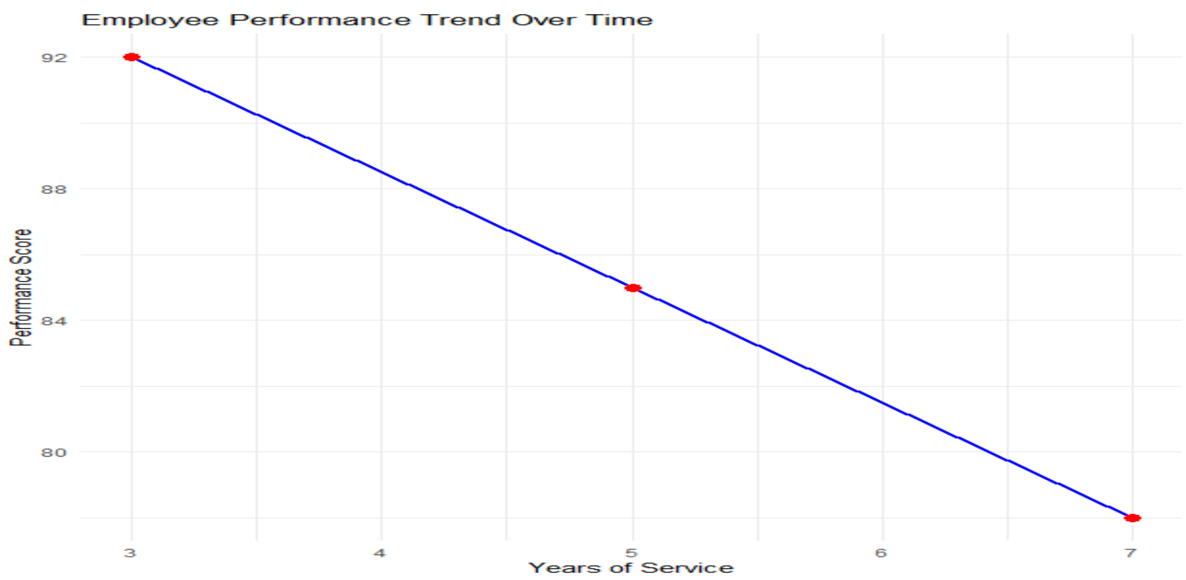
SET 17- Employee Performance Analysis

17A. Line Chart

Program:

```
library(ggplot2)
employee_data <- data.frame(
  EmployeeID = c(1,2,3),
  Department = c("Sales","HR","Marketing"),
  YearsOfService = c(5,3,7),
  PerformanceScore = c(85,92,78)
)
ggplot(employee_data,
       aes(x = YearsOfService, y = PerformanceScore, group = 1)) +
  geom_line(color = "blue", linewidth = 1) +
  geom_point(size = 3, color = "red") +
  labs(
    title = "Employee Performance Trend Over Time",
    x = "Years of Service",
    y = "Performance Score"
  ) +
  theme_minimal()
```

Output:

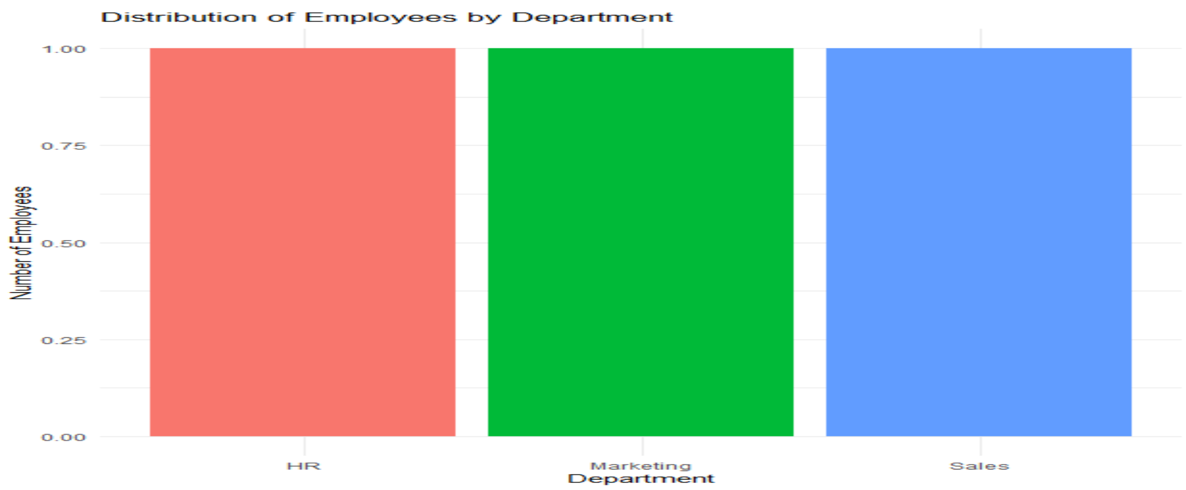


17B. Bar Chart

Program:

```
library(ggplot2)
ggplot(employee_data,
       aes(x = Department, fill = Department)) +
  geom_bar() +
  labs(
    title = "Distribution of Employees by Department",
    x = "Department",
    y = "Number of Employees"  ) +
  theme_minimal() +
  theme(legend.position = "none")
```

Output:



17C. Table

Program:

```
View(employee_data)
```

Output:

Data: employee_data

	EmployeeID	Department	YearsOfService	PerformanceScore
1	1	Sales	5	85
2	2	HR	3	92
3	3	Marketing	7	78

17D. Tableau Dashboard

Program:

```
write.csv(employee_data,
          "Employee_Performance.csv",
          row.names = FALSE)
```

Output:

EmployeeI	Departmer	YearsOfSe	PerformanceScore	
1	Sales	5	85	
2	HR	3	92	
3	Marketing	7	78	

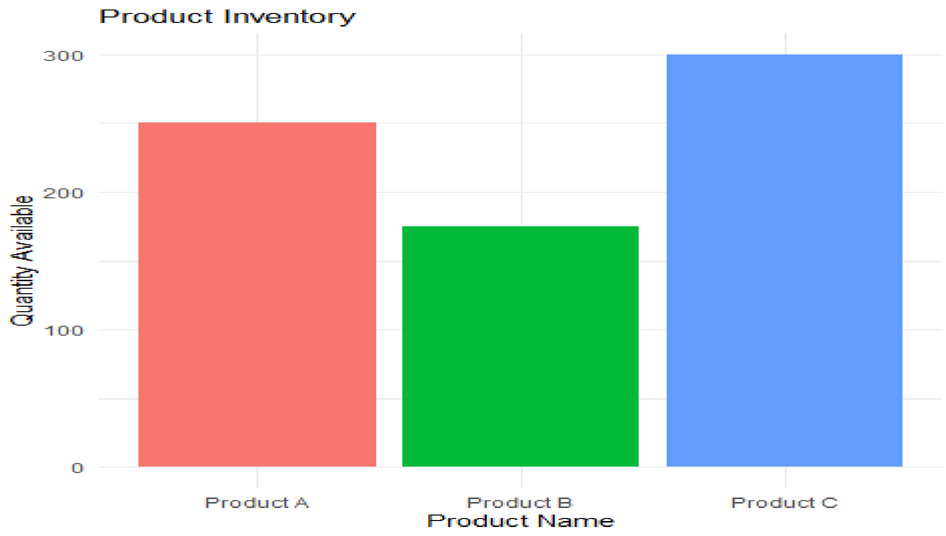
SET 18- Product Inventory Management

18A. Bar Chart

Program:

```
library(ggplot2)
inventory_data <- data.frame(
  ProductID = c(1,2,3),
  ProductName = c("Product A","Product B","Product C"),
  QuantityAvailable = c(250,175,300),
  Price = c(20,15,18))
ggplot(inventory_data,
       aes(x = ProductName, y = QuantityAvailable, fill = ProductName)) +
  geom_bar(stat = "identity") +
  labs(
    title = "Quantity of Products in Inventory",
    x = "Product Name",
    y = "Quantity Available"
  ) +
  theme_minimal() +
  theme(legend.position = "none")
```

Output:

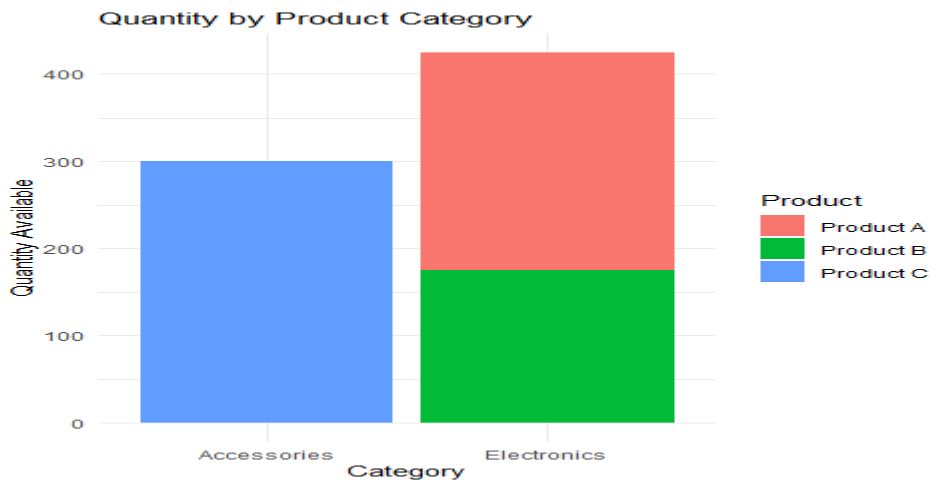


18B. Stacked Bar Chart

Program:

```
library(ggplot2)
inventory_data$Category <- c("Category 1","Category 1","Category 2")
ggplot(inventory_data,
       aes(x = Category, y = QuantityAvailable, fill = ProductName)) +
  geom_bar(stat = "identity") +  labs(
    title = "Quantity by Product Category",
    x = "Product Category",
    y = "Quantity Available"  ) +
  theme_minimal()
```

Output:



18C. Bar Chart

Program:

```
View(inventory_data)
```

Output:

Show 10 entries

Search:

Product Inventory Data

	ProductID	ProductName	QuantityAvailable	Price	Category
1	1	Product A	250	20	Electronics
2	2	Product B	175	15	Electronics
3	3	Product C	300	18	Accessories

Showing 1 to 3 of 3 entries

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18D. Tableau Dashboard

Program:

```
write.csv(
  inventory_data,
  "Product_Inventory.csv",
  row.names = FALSE
)
```

Output:

ProductID	ProductNa	QuantityAv	Price	Category
1	Product A	250	20	Category 1
2	Product B	175	15	Category 1
3	Product C	300	18	Category 2

SET 19- Survey Responses Analysis

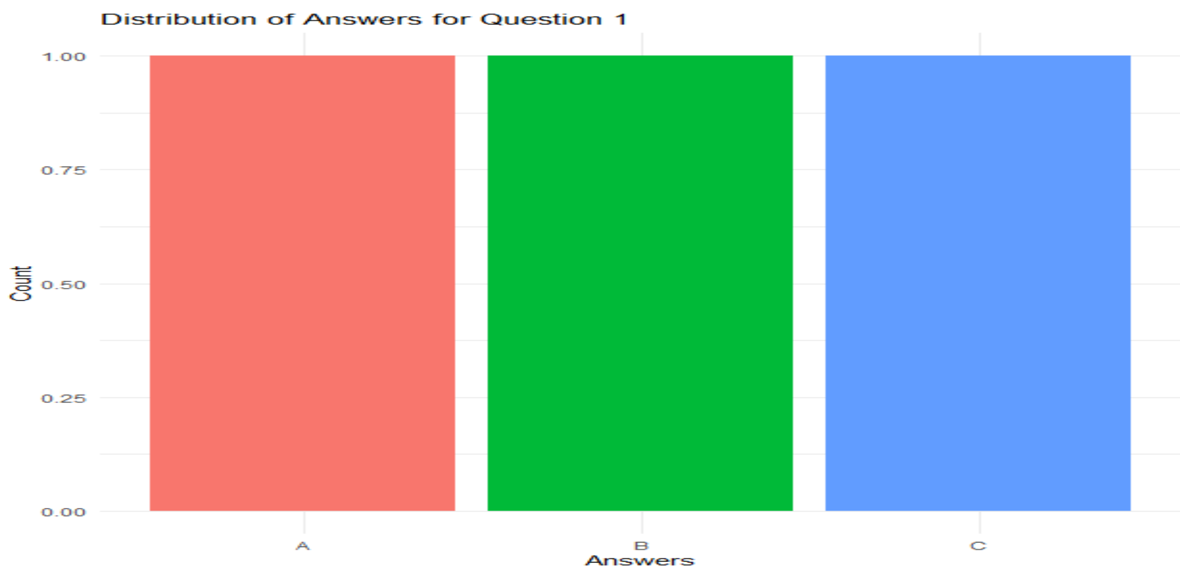
19A. Grouped Bar Chart

Program:

```
library(ggplot2)
survey_data <- data.frame(
  SurveyID = c(1,2,3),
  Question1 = c("A","B","C"),
```

```
  Question2 = c("B","A","A"),
  Question3 = c("C","D","B")
)
q1_data <- as.data.frame(table(survey_data$Question1))
colnames(q1_data) <- c("Answer","Count")
ggplot(q1_data, aes(x = Answer, y = Count, fill = Answer)) +
  geom_bar(stat = "identity") +
  labs(
    title = "Distribution of Answers for Question 1",
    x = "Answers",
    y = "Count"
  ) +
  theme_minimal() +
  theme(legend.position = "none")
```

Output:



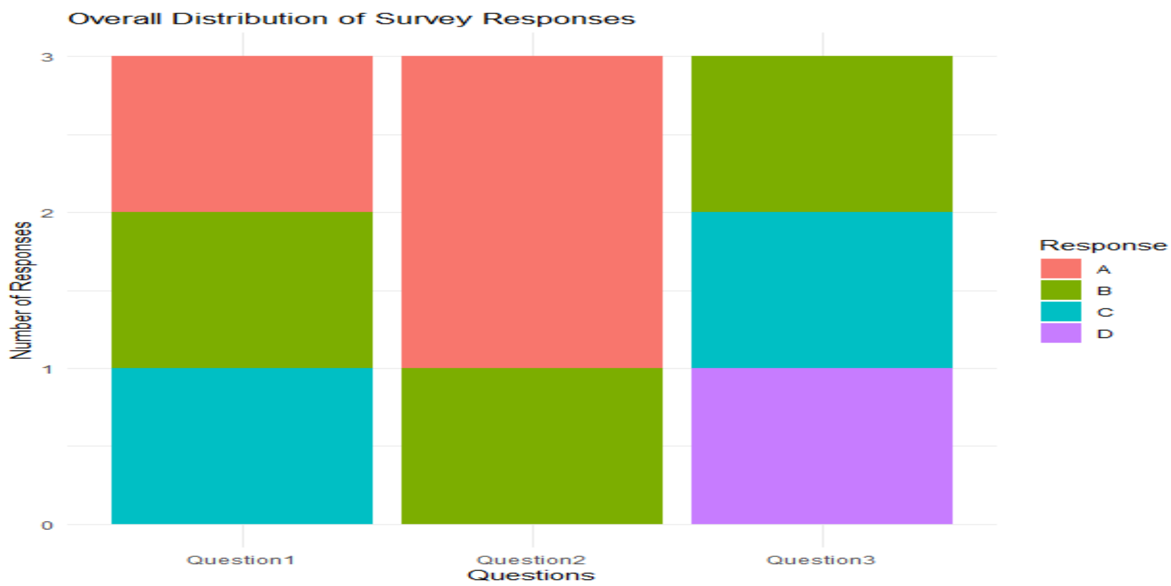
19B. Stacked Bar Chart

Program:

```
library(ggplot2)
library(tidyr)
survey_long <- pivot_longer(
```

```
  survey_data,
  cols = starts_with("Question"),
  names_to = "Question",
  values_to = "Response"
)
ggplot(survey_long,
       aes(x = Question, fill = Response)) +
  geom_bar() +
  labs(
    title = "Overall Distribution of Survey Responses",
    x = "Questions",
    y = "Number of Responses"
  ) +
  theme_minimal()library(ggplot2)
library(tidyr)
survey_long <- pivot_longer(
  survey_data,
  cols = starts_with("Question"),
  names_to = "Question",
  values_to = "Response"
)
ggplot(survey_long,
       aes(x = Question, fill = Response)) +
  geom_bar() +
  labs(
    title = "Overall Distribution of Survey Responses",
    x = "Questions",
    y = "Number of Responses"
  ) +
  theme_minimal()
```

Output:



19C. Table

Program:

View(survey_data)

Output:

Data: survey_data

	SurveyID	Question1	Question2	Question3
1	1	A	B	C
2	2	B	A	D
3	3	C	A	B

19D. Tableau Dashboard

Program:

```
write.csv(
  survey_data,
  "Survey_Responses.csv",
  row.names = FALSE
)
```

Output:

SurveyID	Question1	Question2	Question3
1	A	B	C
2	B	A	D
3	C	A	B

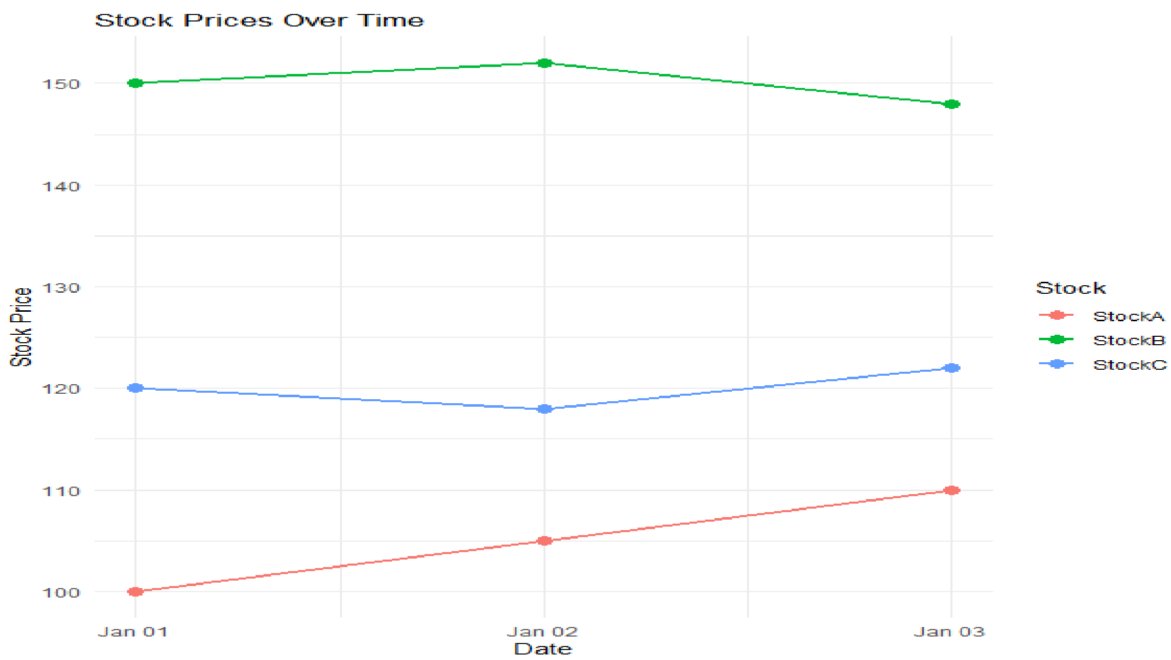
SET 20- Stock Analysis

20A. Line Chart

Program:

```
library(ggplot2)
library(tidyr)
stock_data <- data.frame(
  Date = as.Date(c("2023-01-01","2023-01-02","2023-01-03")),
  StockA = c(100,105,110),
  StockB = c(150,152,148),
  StockC = c(120,118,122)
)
stock_long <- pivot_longer(
  stock_data,
  cols = c(StockA, StockB, StockC),
  names_to = "Stock",
  values_to = "Price"
)
ggplot(stock_long,
       aes(x = Date, y = Price, color = Stock, group = Stock)) +
  geom_line(linewidth = 1) +
  geom_point(size = 3) +
  labs(
    title = "Stock Prices Over Time",
    x = "Date",
    y = "Stock Price"
  ) +
  theme_minimal()
```

Output:

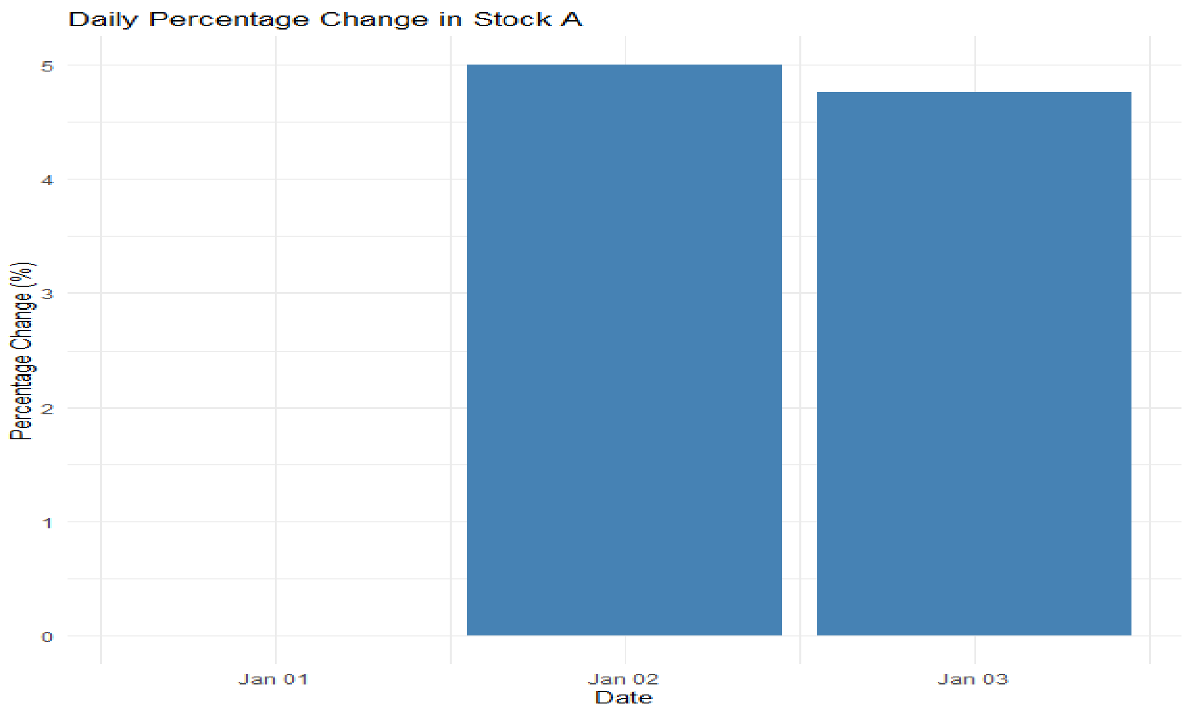


20B. Bar Chart

Program:

```
library(ggplot2)
stock_data$PercentageChange <- c(
  0,
  (105 - 100) / 100 * 100,
  (110 - 105) / 105 * 100
)
ggplot(stock_data,
       aes(x = Date, y = PercentageChange)) +
  geom_bar(stat = "identity", fill = "steelblue") +
  labs(
    title = "Daily Percentage Change in Stock A",
    x = "Date",
    y = "Percentage Change (%)"
  ) +
  theme_minimal()
```

Output:



20C. Table

Program:

```
View(stock_data)
```

Output:

Data: stock_data

	Date	StockA	StockB	StockC	PercentageChange
1	2023-01-01	100	150	120	0.000000
2	2023-01-02	105	152	118	5.000000
3	2023-01-03	110	148	122	4.761905

20D. Tableau Dashboard

Program:

```
write.csv(
  stock_data,
  "Stock_Prices.csv",
  row.names = FALSE
)
```

Output:

Date	StockA	StockB	StockC	PercentageChange
########	100	150	120	0
########	105	152	118	5
########	110	148	122	4.761905